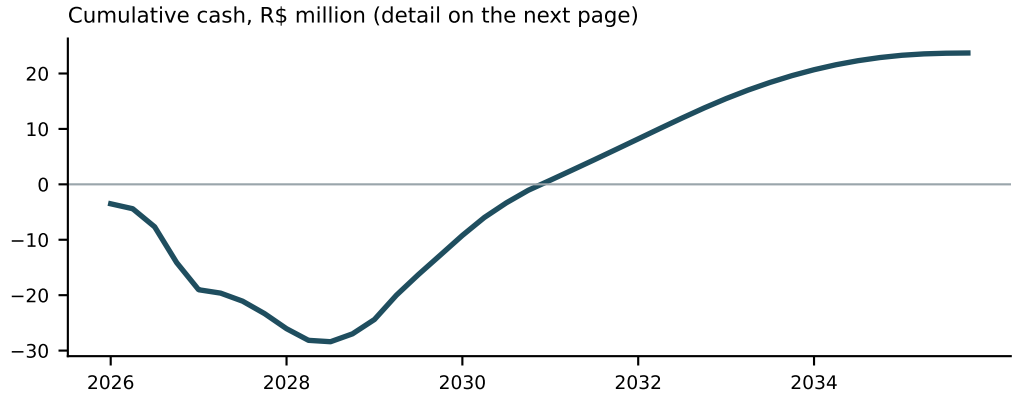
# Land portfolio: cost x return - executive summary

Synthetic data generated by generate_data.py. Every number is computed by the model in this repository.

| Portfolio | Base case |
|---|---|
| Parcels | 10 |
| VGV (total sales value) | R$ 59.5 mi |
| subdivided / sold whole | R$ 52.2 mi / R$ 7.3 mi |
| Net profit (nominal) | R$ 23.7 mi |
| Margin on VGV | 39.8% |
| NPV at 14% a year | R$ 2.2 mi |
| IRR per year | 16.6% |
| Maximum cash exposure | R$ 28.4 mi (Sep 2028) |
| Payback | Feb 2031 |

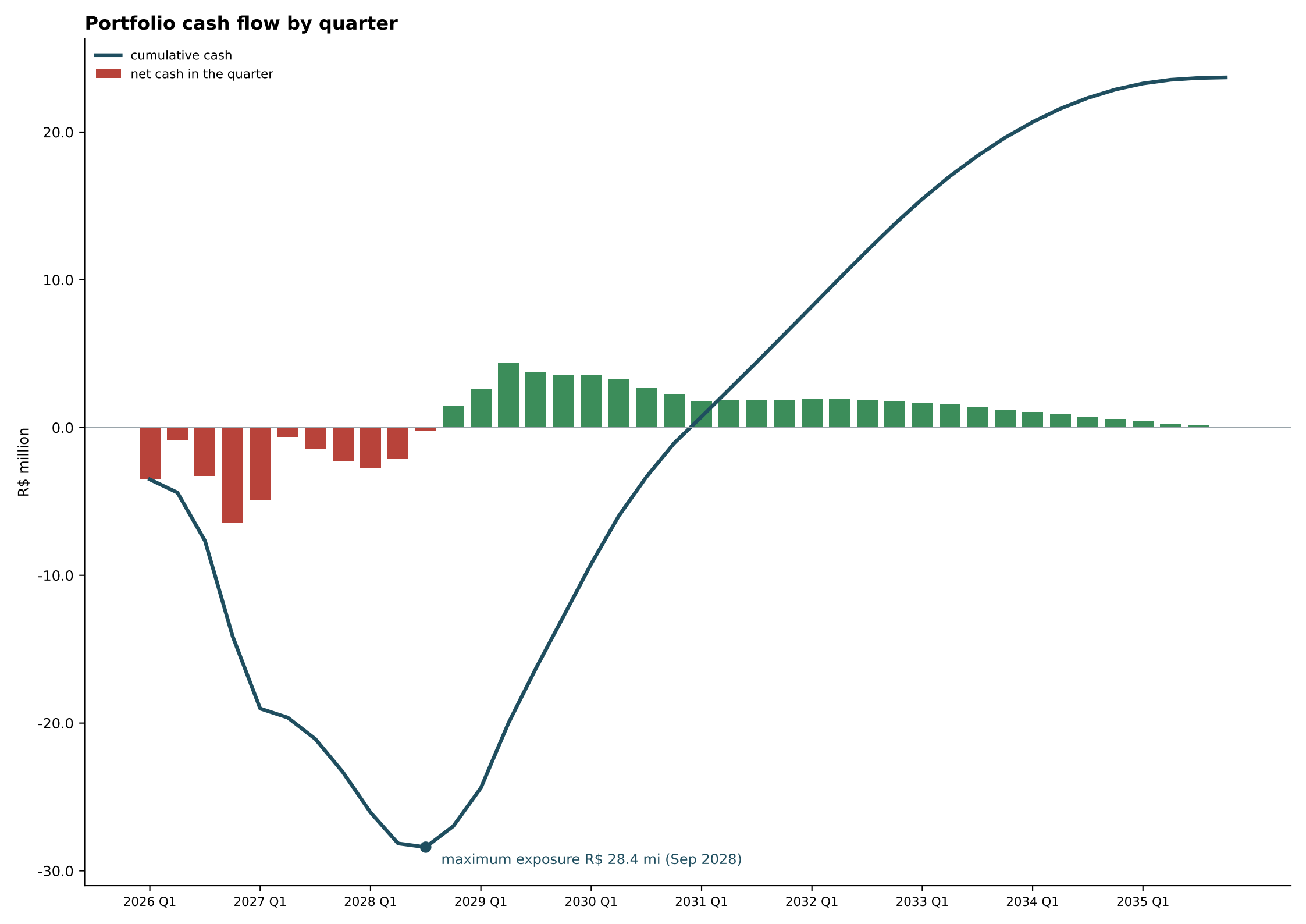
| Scenario | VGV | NPV | IRR | Max exposure |
|---|---|---|---|---|
| pessimistic | R$ 50.4 mi | R$ -11.8 mi | 2.8% | R$ 39.9 mi |
| base | R$ 59.5 mi | R$ 2.2 mi | 16.6% | R$ 28.4 mi |
| optimistic | R$ 65.6 mi | R$ 12.3 mi | 31.5% | R$ 20.4 mi |

Pessimistic and optimistic move lot prices, sales speed, infrastructure cost, holding time, appreciation and default to the ends of their ranges.




## What the numbers say

- The 4 subdivisions carry the portfolio: NPV of R$ 3.2 million against R$ 2.2 million in total.
- The 6 parcels sold whole return 4.1% to 10.9% a year; 6 of them are below the 14% hurdle, because holding land costs maintenance and tax while it appreciates more slowly than the hurdle rate.
- Infrastructure cost moves NPV the most: from R$ 5.9 million at R$ 80/m2 to R$ -4.0 million at R$ 160/m2.
- Cash need peaks at R$ 28.4 million in Sep 2028; the portfolio pays back in Feb 2031, 61 months after the first purchase.
- In the pessimistic scenario NPV is R$ -11.8 million and IRR 2.8%; the optimistic one reaches R$ 12.3 million.
- Subdividing beats selling whole in 4 of the 4 urban parcels where both are possible.
- AI memos (numbers checked against the model): 6 do not invest, 4 invest with conditions.


Portfolio cash flow by quarter
cumulative cash
net cash in the quarter
R$ million
20.0
10.0
0.0
-10.0
-20.0
-30.0
maximum exposure R$ 28.4 mi (Sep 2028)
2026 Q1
2027 Q1
2028 Q1
2029 Q1
2030 Q1
2031 Q1
2032 Q1
2033 Q1
2034 Q1
2035 Q1

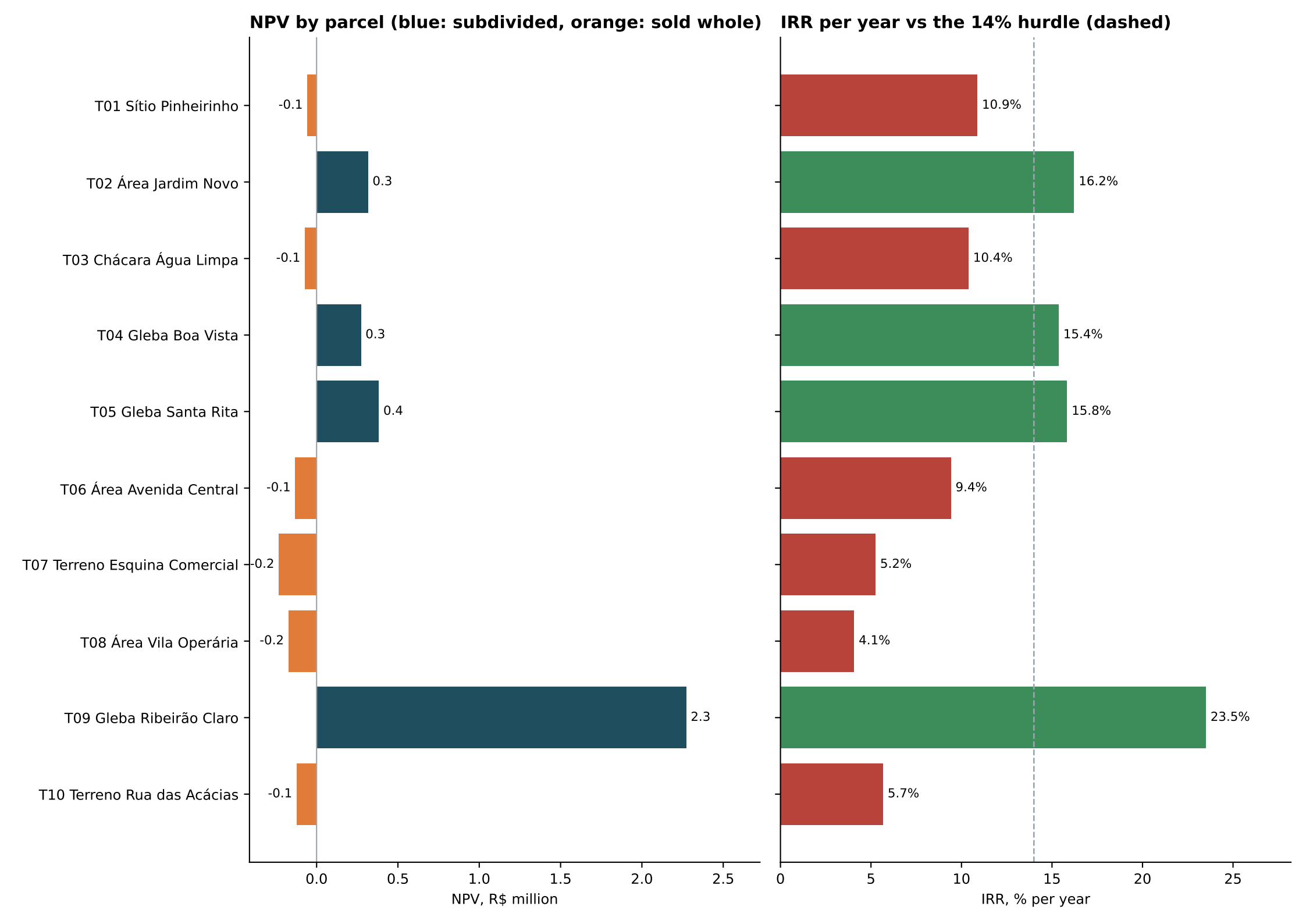
# Parcels

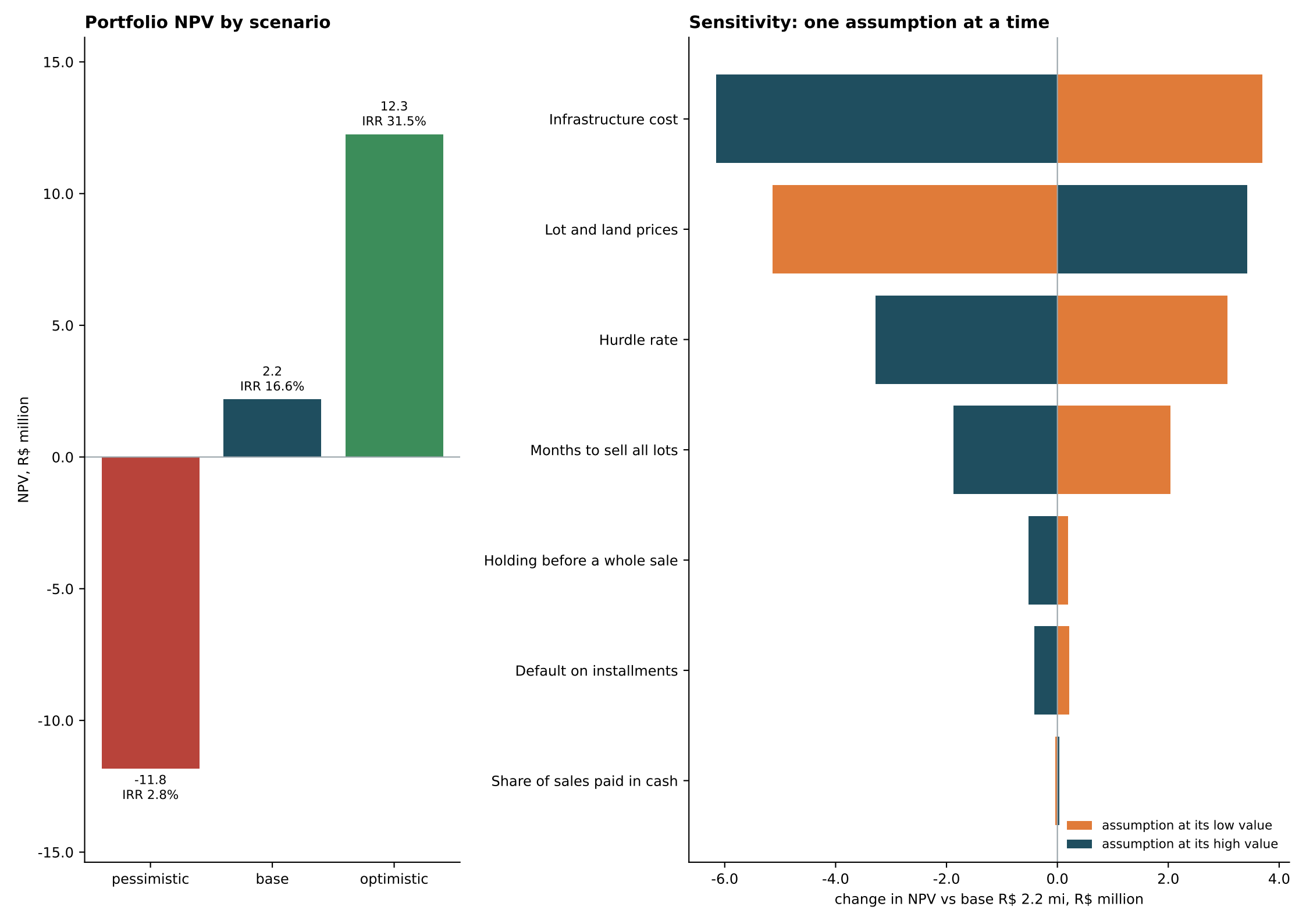
Hurdle rate 14% a year. Payback in months after purchase.

| Id | Parcel | Kind | Area m2 | Strategy | Price | VGV | NPV | IRR | Max exposure | Payback | AI memo |
|---|---|---|---|---|---|---|---|---|---|---|---|
| T01 | Sítio Pinheirinho | rural | 54,000 | whole | R$ 693 k | R$ 1.1 mi | R$ -57 k | 10.9% | R$ 764 k | 38 | do not invest |
| T02 | Área Jardim Novo | urban | 33,800 | subdivision | R$ 2.6 mi | R$ 8.6 mi | R$ 315 k | 16.2% | R$ 4.3 mi | 56 | invest with conditions |
| T03 | Chácara Água Limpa | rural | 75,500 | whole | R$ 755 k | R$ 1.2 mi | R$ -71 k | 10.4% | R$ 842 k | 38 | do not invest |
| T04 | Gleba Boa Vista | expansion | 68,000 | subdivision | R$ 3.0 mi | R$ 12.6 mi | R$ 272 k | 15.4% | R$ 6.8 mi | 59 | invest with conditions |
| T05 | Gleba Santa Rita | expansion | 73,600 | subdivision | R$ 3.0 mi | R$ 13.4 mi | R$ 382 k | 15.8% | R$ 7.1 mi | 59 | invest with conditions |
| T06 | Área Avenida Central | urban | 10,000 | whole | R$ 1.1 mi | R$ 1.7 mi | R$ -131 k | 9.4% | R$ 1.2 mi | 38 | do not invest |
| T07 | Terreno Esquina Comercial | urban | 4,000 | whole | R$ 1.1 mi | R$ 1.5 mi | R$ -232 k | 5.2% | R$ 1.2 mi | 40 | do not invest |
| T08 | Área Vila Operária | urban | 7,700 | whole | R$ 705 k | R$ 953 k | R$ -172 k | 4.1% | R$ 773 k | 41 | do not invest |
| T09 | Gleba Ribeirão Claro | expansion | 76,200 | subdivision | R$ 3.6 mi | R$ 17.6 mi | R$ 2.3 mi | 23.5% | R$ 7.3 mi | 49 | invest with conditions |
| T10 | Terreno Rua das Acácias | urban | 2,600 | whole | R$ 590 k | R$ 821 k | R$ -122 k | 5.7% | R$ 638 k | 40 | do not invest |

## Subdivide or sell whole (urban parcels where both are possible)

| Id | Parcel | Area m2 | Rule picks | NPV subdivided | IRR | NPV sold whole | IRR |
|---|---|---|---|---|---|---|---|
| T02 | Área Jardim Novo | 33,800 | subdivision | R$ 315 k | 16.2% | R$ -573 k | 5.3% |
| T04 | Gleba Boa Vista | 68,000 | subdivision | R$ 272 k | 15.4% | R$ -590 k | 6.1% |
| T05 | Gleba Santa Rita | 73,600 | subdivision | R$ 382 k | 15.8% | R$ -336 k | 9.7% |
| T09 | Gleba Ribeirão Claro | 76,200 | subdivision | R$ 2.3 mi | 23.5% | R$ -322 k | 10.6% |


NPV by parcel (blue: subdivided, orange: sold whole)
IRR per year vs the 14% hurdle (dashed)
T01 Sítio Pinheirinho
T02 Área Jardim Novo
T03 Chácara Água Limpa
T04 Gleba Boa Vista
T05 Gleba Santa Rita
T06 Área Avenida Central
T07 Terreno Esquina Comercial
T08 Área Vila Operária
T09 Gleba Ribeirão Claro
T10 Terreno Rua das Acácias
-0.1
0.3
-0.1
0.3
0.4
-0.1
-0.2
-0.2
2.3
-0.1
10.9%
16.2%
10.4%
15.4%
15.8%
9.4%
5.2%
4.1%
23.5%
5.7%
0.0
0.5
1.0
1.5
2.0
2.5
NPV, R$ million
0
5
10
15
20
25
IRR, % per year


Portfolio NPV by scenario
NPV, R$ million
15.0
10.0
5.0
0.0
-5.0
-10.0
-15.0
12.3
IRR 31.5%
2.2
IRR 16.6%
-11.8
IRR 2.8%
pessimistic
base
optimistic
Sensitivity: one assumption at a time
Infrastructure cost
Lot and land prices
Hurdle rate
Months to sell all lots
Holding before a whole sale
Default on installments
Share of sales paid in cash
assumption at its low value
assumption at its high value
-6.0
-4.0
-2.0
0.0
2.0
4.0
change in NPV vs base R$ 2.2 mi, R$ million

# Key assumptions

Base value and the range used in scenarios and sensitivity. 'illustrative' = typical value, not market research. Full list: sheet Assumptions.

| Assumption | Base | Range | Unit | Source |
|---|---|---|---|---|
| infra_per_m2 | 110 | 80 - 160 | R$/m2 saleable | streets, drainage, water, sewage, power (illustrative) |
| lot_size_m2 | 300 | 250 - 360 | m2 | illustrative |
| saleable_ratio | 0.6 | 0.55 - 0.65 | share of gross area | streets and public areas are set by municipal law under Lei 6.766/1979 (illustrative) |
| approval_months | 12 | 8 - 24 | months | illustrative |
| absorption_months | 36 | 24 - 48 | months to sell all lots | author's experience: return takes 2-4 years on average |
| itbi_rate | 0.03 | 0.02 - 0.03 | share of price | municipal transfer tax (Constitution art. 156, II); 2-3% is the common rate (illustrative) |
| deed_registry_rate | 0.015 | 0.01 - 0.02 | share of price | deed and registry fees follow state fee tables (in SP, Lei 11.331/2002); 1-2% of the [...] |
| iptu_rate | 0.01 | 0.005 - 0.02 | share of land value per year | municipal urban property tax; vacant land often pays more; CTN art. 32 extends it to [...] |
| itr_rate | 0.003 | 0.0003 - 0.01 | share of bare-land value per yea | rural land tax, Lei 9.393/1996: 0.03% to 1.00% for areas up to 50 ha, by degree of use; [...] |
| sales_tax_rate | 0.0673 | 0.0593 - 0.0673 | share of receipts | Lucro Presumido on cash receipts: IRPJ 15% + 10% surcharge and CSLL 9% on presumed profit [...] |
| broker_rate | 0.05 | 0.04 - 0.06 | share of sale price | illustrative |
| installments | 60 | 36 - 120 | months | illustrative |
| monthly_interest | 0.008 | 0.005 - 0.01 | per month, Price table | illustrative |
| index_rate_year | 0.045 | 0.035 - 0.06 | per year, fixed | correction index (IPCA by default) as a fixed assumption, no real series |
| default_rate | 0.01 | 0 - 0.03 | share of installments never rece | author's experience: delinquency is rare |
| discount_rate_year | 0.14 | 0.11 - 0.18 | per year | hurdle rate; compare with the Selic of the day (illustrative) |

# Investment memos

Written by Claude from the model's facts; every number was checked against those facts before the memo was kept.

## T01 Sítio Pinheirinho - do not invest

- Sítio Pinheirinho does not clear our 14.0% hurdle rate, so we should pass.
- The whole-parcel strategy shows a net profit of R$ 254,823 on a VGV of R$ 1,124,426. That is a 22.7% margin. But the NPV at the 14.0% hurdle rate is R$ -56,549, and the IRR is only 10.9% per year. Capital is tied up for 43 months, and payback takes 38 months. The deal earns less than our required return.
- Risk: IRR of 10.9% per year sits below the 14.0% hurdle rate.
- Risk: Long cash flow duration of 43 months and maximum cash exposure of R$ 764,282 tie up capital.
- Risk: Holding costs of R$ 37,598 for maintenance and property tax erode returns over the hold.
- Condition: Purchase price falls enough to bring NPV at the 14.0% hurdle rate to zero or above.
- Condition: Sales timeline shortens materially, so the IRR clears 14.0% per year.

## T02 Área Jardim Novo - invest with conditions

- T02 Área Jardim Novo clears the hurdle rate, but payback at 56 months is long and the case needs firm sales pricing.
- The parcel shows a positive NPV of R$ 315,229 at the 14.0% hurdle rate and an IRR of 16.2% per year. Net profit is R$ 3,572,060 on a VGV of R$ 8,582,700, a margin of 41.6%. Maximum cash exposure reaches R$ 4,328,382, and the cash flow runs for 108 months. The margin is healthy, but the return depends on selling 67 lots at R$ 128,100 each.
- Risk: Payback at 56 months ties up capital for a long time.
- Risk: Infrastructure and site labor of R$ 2,387,880 could overrun.
- Risk: NPV is only R$ 315,229, so a small drop in prices or sales pace erases the hurdle margin.
- Condition: Confirm the R$ 128,100 price per lot with pre-sale demand before committing.
- Condition: Lock a fixed-price construction contract for infrastructure and site labor.

## T03 Chácara Água Limpa - do not invest

- T03 Chácara Água Limpa does not clear our 14.0% hurdle rate, so we should pass.
- The parcel shows a nominal net profit of R$ 265,787 on a VGV of R$ 1,222,756, a margin of 21.7%. But the NPV at the 14.0% hurdle rate is R$ -71,376, and the IRR is only 10.4% per year. Payback takes 38 months within a 43-month cash flow, and maximum cash exposure reaches R$ 842,438.
- Risk: Returns fall below the 14.0% hurdle rate, so the deal destroys value on a risk-adjusted basis.
- Risk: Long payback of 38 months ties up capital for most of the 43-month cash flow duration.
- Risk: Holding costs of R$ 50,962 for maintenance and property tax erode the profit.
- Condition: Acquisition price or sales assumptions would need to improve enough to bring NPV at 14.0% to zero or above.

# Investment memos

Written by Claude from the model's facts; every number was checked against those facts before the memo was kept.

## T04 Gleba Boa Vista - invest with conditions

- T04 Gleba Boa Vista clears our 14.0% hurdle rate, but payback is slow, so invest only if the conditions hold.
- The parcel shows a net profit of R$ 4,804,336 on a VGV of R$ 12,566,400, a margin of 38.2%. NPV at the 14.0% hurdle rate is R$ 271,873, and IRR is 15.4% per year. The maximum cash exposure is R$ 6,757,664, and payback takes 59 months.
- Risk: Payback of 59 months within a 108-month cash flow leaves a long period of capital at risk.
- Risk: Infrastructure and site labor of R$ 4,847,040 is a large cost that could overrun.
- Risk: The NPV margin over the hurdle rate is thin, so small changes in price or timing could push returns below 14.0%.
- Condition: Confirm the price per lot of R$ 92,400 across the 136 lots before committing.
- Condition: Lock the infrastructure budget of R$ 4,847,040 with fixed-price contracts.

## T05 Gleba Santa Rita - invest with conditions

- Gleba Santa Rita clears the hurdle rate, but payback is long and cash exposure is high, so proceed only with conditions.
- The parcel shows an NPV of R$ 381,691 at the 14.0% hurdle rate and an IRR of 15.8% per year. Net profit is R$ 5,205,962 on a VGV of R$ 13,406,400, a margin of 38.8%. The subdivision plan yields 147 lots at R$ 91,200 each. Maximum cash exposure reaches R$ 7,129,131, and payback takes 59 months.
- Risk: Payback of 59 months ties up capital for a long period within a 108-month cash flow.
- Risk: Infrastructure and site labor of R$ 5,239,080 is large relative to the acquisition price, so cost overruns would cut returns quickly.
- Risk: NPV is positive but thin, so a small drop in lot prices or sales pace could push it below the hurdle rate.
- Condition: Confirm infrastructure and site labor costs with fixed-price quotes before committing capital.
- Condition: Secure pre-sales or a sales pace that supports the 12-month launch timeline.

## T06 Área Avenida Central - do not invest

- T06 Área Avenida Central destroys value at our 14.0% hurdle rate, so we should pass.
- The parcel shows a net profit of R$ 347,299 on a R$ 1,113,000 acquisition, a 20.1% margin on VGV. But the IRR is 9.4% per year, below the 14.0% hurdle rate. At that hurdle the NPV is R$ -130,838, so the deal loses value after capital cost. Payback takes 38 months over a 43-month cash flow.
- Risk: Returns fall short of the hurdle: IRR of 9.4% per year against 14.0%.
- Risk: Long capital tie-up: maximum cash exposure of R$ 1,214,910 for up to 43 months.
- Risk: Holding costs of R$ 51,825 erode the margin over the cash flow period.
- Condition: Acquisition price is renegotiated low enough to lift NPV at the 14.0% hurdle rate to zero or above.

# Investment memos

Written by Claude from the model's facts; every number was checked against those facts before the memo was kept.

## T07 Terreno Esquina Comercial - do not invest

- T07 Terreno Esquina Comercial clears no hurdle: NPV at 14.0% is R$ -231,538, so we should pass.
- The parcel costs R$ 1,074,000 and sells for a VGV of R$ 1,476,098. Net profit is R$ 179,058, a margin of 12.1% on VGV. The IRR is only 5.2% per year, well below the 14.0% hurdle rate. Payback takes 40 months over a 43-month cash flow, which ties up capital for too long.
- Risk: Returns fall far short of the 14.0% hurdle rate, with an IRR of 5.2% per year.
- Risk: Maximum cash exposure reaches R$ 1,158,780, and holding costs of R$ 36,450 erode the margin.
- Risk: Payback at 40 months leaves little room for any sales delay.
- Condition: Acquisition price falls enough to bring NPV at 14.0% to zero or better.
- Condition: Sales value rises or holding costs fall enough to lift the IRR well above 14.0%.

## T08 Área Vila Operária - do not invest

- T08 Área Vila Operária fails the 14.0% hurdle rate, so we should pass.
- The parcel shows a nominal net profit of R$ 90,998 on a VGV of R$ 953,102, a margin of 9.5%. At the 14.0% hurdle rate the NPV is R$ -172,327, and the IRR is only 4.1% per year. Payback takes 41 months, and maximum cash exposure reaches R$ 772,830.
- Risk: Returns are far below the 14.0% hurdle rate, so capital is destroyed in present-value terms.
- Risk: Payback at 41 months within a 43-month cash flow leaves little room for delays in sales or permits.
- Risk: Holding costs of R$ 36,105 add to the drag on an already thin margin.
- Condition: Acquisition price or total sales value would need to change enough to bring NPV at the 14.0% hurdle rate to zero or above.

## T09 Gleba Ribeirão Claro - invest with conditions

- Parcel T09 Gleba Ribeirão Claro clears the 14.0% hurdle with an NPV of R$ 2,271,440, but cash exposure and a 49-month payback require conditions.
- The subdivision of 76,200 m2 into 152 lots at R$ 115,800 per lot gives a VGV of R$ 17,601,600 and a net profit of R$ 8,874,451. The margin on VGV is 50.4%, and the IRR is 23.5% per year against a 14.0% hurdle. The purchase price is R$ 3,563,000, but maximum cash exposure reaches R$ 7,335,669.
- Risk: Maximum cash exposure of R$ 7,335,669 is more than twice the acquisition price.
- Risk: Payback takes 49 months within a 108-month cash flow duration, so returns depend on sustained sales.
- Risk: Infrastructure and site labor of R$ 5,417,280 is a large cost that could overrun.
- Condition: Confirm the infrastructure and site labor budget before committing capital.
- Condition: Secure financing or equity commitments that cover the full maximum cash exposure of R$ 7,335,669.

# Investment memos

Written by Claude from the model's facts; every number was checked against those facts before the memo was kept.

## T10 Terreno Rua das Acácias - do not invest

- T10 Terreno Rua das Acácias does not clear our 14.0% hurdle rate, so we should pass on it.
- The parcel shows a nominal net profit of R$ 106,659, a margin on VGV of 13.0%, and an IRR of 5.7% per year. At the 14.0% hurdle rate, the NPV is R$ -121,642, which means the deal destroys value for us. Payback takes 40 months over a 43-month cash flow, which ties up capital for too long.
- Risk: The IRR of 5.7% per year sits far below the 14.0% hurdle rate, so returns would not compensate for the capital at risk.
- Risk: Maximum cash exposure reaches R$ 637,540, and holding costs of R$ 20,990 keep eating into the margin over the long timeline.
- Condition: The acquisition price or sales value would need to change enough to bring the NPV at the 14.0% hurdle rate to zero or above.